Advanced Data Analytics & Machine Learning

Data Auditing & Missingness Mechanisms in MOH Records – Week 1, Day 3

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Abstract

This session investigates **missing data mechanisms** and **imputation strategies** in a real Ministry of Health (MOH) facility dataset of 4,999 patient records.

- **Missingness audit:** quantifying and visualising missing values across BMI, hemoglobin, creatinine, HIV status, and marital status.
- **Mechanism classification:** MCAR, MAR, and MNAR — with statistical tests (t-test, chi-square, Fisher's exact) to assess each.
- **Data leakage prevention:** the critical rule of splitting *before* imputation and building correct preprocessing workflows.
- **Four imputation strategies compared:** Listwise Deletion, Median, KNN, and MICE — evaluated via Logistic Regression on a hypertension outcome.
- **Evidence-based recommendation:** KNN achieves the highest AUC (0.772) while retaining the full sample; MICE preferred for statistical inference.

Philosophy: *classify the mechanism first, then match the treatment — never impute blindly.*

Session Objectives

- **Audit** a national MOH hypertension dataset for missing clinical and laboratory values.
- **Classify** the missingness mechanism (MCAR, MAR, MNAR) for each affected variable using statistical evidence.
- **Demonstrate** the data leakage risk and apply the correct split-before-imputation workflow.
- **Compare** four imputation strategies — Listwise Deletion, Median, KNN, and MICE — on a real predictive modelling task.
- **Interpret** model performance (Accuracy and AUC-ROC) to recommend the best missing-data treatment.
- **Articulate** best-practice guidelines for each method, including when deletion is safe and when model-based imputation is essential.

Agenda

1. **Part 1: Data Loading & Exploration** (MOH dataset overview, missingness quantification, heatmap)
2. **Part 2: Missingness Mechanisms** (MCAR, MAR, MNAR — definitions, MOH examples, statistical tests)
3. **Part 3: Preventing Data Leakage** (split-before-imputation rule, correct workflow, pipeline principle)
4. **Part 4: Imputation Strategies Compared** (Listwise, Median, KNN, MICE — model evaluation)
5. **Part 5: Best Practices & Recommendations** (method-by-method guide, overall recommendation matrix)

Dataset

MOH hypertension facility dataset (`htn_dat.csv`): 4,999 patients, 16 variables including vitals (SBP, DBP), BMI, hemoglobin, creatinine, HIV status, clinic type, and survival outcomes.

Transition

Part 1: Data Loading & Exploration

"You cannot fix what you cannot see — audit the missingness landscape first."

MOH Facility Dataset: Overview

Key characteristics

- **4,999 patient records** from MOH facilities.
- **16 variables**: demographics, vitals, labs, outcomes.
- **Target variable:** `SBP_ge120` (systolic BP $\geq$ 120).
- Includes IPW weights, survival time, event indicator.

Key variables for analysis

- **Vitals:** SBP, DBP, BMI
- **Labs:** Hemoglobin (centered), Creatinine (log, centered)
- **Demographics:** Age, Gender, Marital status
- **Clinical:** HIV status, ARV-naive, Clinic type

Why This Dataset?

Real-world MOH records are *never* complete. Rural clinics lack equipment; critically ill patients miss labs. This dataset mirrors those realities — making it ideal for practising principled missing-data handling.

Prediction Task

Classify whether a patient has SBP $\geq$ 120 mmHg (hypertension indicator) using age, DBP, BMI, hemoglobin, and creatinine.

Quantifying Missingness: Counts by Column

Missing value counts

Variable	Missing	% Missing
`adv_HIV`	1,961	39.2%
`log_creat_centered`	1,538	30.8%
`hgb_centered`	1,394	27.9%
`BMI`	670	13.4%
`married`	164	3.3%

Scale of the Problem

Over 39% of HIV status and 30% of creatinine values are missing — naive deletion would discard most of the dataset.

Interpretation

- **adv_HIV (39.2%):** likely MNAR — critically ill patients may die before testing.
- **Creatinine (30.8%):** suspected MNAR — sickest patients cannot travel to central labs.
- **Hemoglobin (27.9%):** likely MAR — rural clinics lack centrifuges.
- **BMI (13.4%):** possibly MCAR — random equipment failure.
- **Married (3.3%):** low missingness, likely MCAR.

Visualizing the Missingness Landscape

Missing Data Heatmap

A binary heatmap (missing = 1, observed = 0) across all 4,999 records reveals:

- **Vertical bands** (columns): variables with high missingness (HIV, creatinine, hemoglobin).
- **Horizontal patterns**: whether the same patients are missing multiple values.
- **Random scatter**: suggests MCAR for a given variable.
- **Clustered blocks**: suggests MAR or MNAR.

What the Heatmap Reveals

BMI missingness appears randomly scattered; hemoglobin and creatinine show structured patterns consistent with clinic-level or severity-level drivers.

Reading the Heatmap

- **Dense dark column** $\rightarrow$ high missingness (adv_HIV).
- **Sparse dark pixels** $\rightarrow$ low missingness (married).
- **Co-occurring gaps** in the same rows $\rightarrow$ patients with multiple missing labs (possibly the sickest).

Next Step

Visual inspection generates hypotheses. Statistical tests (t-test, chi-square) confirm or reject them.

Part 2: Missingness Mechanisms

"Before you impute, diagnose why the data are missing."

Missingness Mechanisms: The Decision Framework

The mechanism determines the safe treatment. Misclassification leads to biased estimates and invalid inference.

Mechanism	Definition	MOH Example	Safe Treatment
MCAR Missing Completely At Random	Missingness independent of *all* data (observed and unobserved)	Random power outage kills the weighing scale	Deletion or simple imputation
MAR Missing At Random	Missingness depends on *observed* data only	Rural clinics lack centrifuges $\rightarrow$ missing hemoglobin	Model-based imputation (KNN, MICE)
MNAR Missing Not At Random	Missingness depends on the *unobserved* value itself	Critically ill patients too sick for creatinine testing	Sensitivity analysis; domain model

Hierarchy of Difficulty

MCAR is easiest to handle but rarest in practice. MNAR is most common in clinical data but hardest to prove.

MCAR: Missing Completely At Random

Definition
The probability that a value is missing is *completely unrelated* to any patient characteristic — whether observed or unobserved.

Key properties

- Missingness is pure random noise.
- The strongest and most restrictive assumption.
- If true, complete-case analysis yields unbiased estimates.
- Very rare in clinical and epidemiological data.

Statistical assessment
Compare observed characteristics (e.g., age, sex) between patients *with* and *without* missing values. No systematic difference → consistent with MCAR.

MOH Example

A random power outage at a rural clinic causes the digital weighing scale to fail. BMI measurements are missing for a *random* subset of patients — unrelated to their weight, age, or health status.

Critical Warning

MCAR cannot be "proven" — you can only fail to reject it. A non-significant test does not guarantee MCAR; it merely says you found no evidence against it.

MCAR Assessment: Testing BMI Missingness

Approach

Compare the mean **age** of patients with missing BMI versus those with observed BMI using an independent-samples t-test.

Hypotheses

- H_0: Mean age is the same in both groups (consistent with MCAR).
- H_1: Mean age differs (BMI missingness is NOT MCAR).

Logic

If missingness is truly random, the two groups should be demographically similar. A significant difference implies systematic missingness.

Result

T-test p-value = 0.3517

Since $p = 0.3517 > 0.05$:

- Fail to reject H_0.
- No significant age difference between groups.
- BMI missingness *may be* MCAR.

Interpretation

There is no evidence that BMI missingness is related to patient age. The pattern is consistent with random equipment failure — supporting the MCAR assumption.

MCAR: Implications for Treatment

If MCAR holds:

- Complete-case (listwise) deletion produces *unbiased* estimates.
- Simple imputation (mean/median) is also valid.
- The main cost is *reduced sample size* and statistical power.
- No selection bias is introduced by removing incomplete rows.

For our BMI variable:

- 670 records (13.4%) have missing BMI.
- Deletion is *acceptable* if sample size permits.
- Median imputation is a safe, simple alternative.

Key Point

MCAR is the *only* mechanism where listwise deletion is guaranteed to be unbiased. This is why correctly identifying MCAR matters — it justifies the simplest treatment.

Caveat

We tested only age. Other unmeasured factors could still drive missingness. MCAR should be treated as a *working assumption*, not a proven fact.

MAR: Missing At Random

Definition

The probability that a value is missing depends on *other observed variables*, but *not* on the missing value itself.

Key properties

- Missingness is *systematic*, but the system is observable.
- More realistic than MCAR for most clinical data.
- Within strata defined by observed variables, missingness is random.
- The foundation assumption for multiple imputation (MICE).

Statistical assessment

Test whether the *proportion* of missing values differs across groups defined by an observed variable (e.g., clinic type).

MOH Example

Hemoglobin (`hgb_centered`) results are missing more often in *rural* clinics because they lack centrifuges. Clinic type (observed) explains the missingness. Within rural clinics, the missingness is assumed random.

"At Random" Is Misleading

MAR does *not* mean the missingness is random in the colloquial sense. It means: *conditional on observed covariates*, the probability of missingness does not depend on the missing value itself.

MAR Assessment: Hemoglobin by Clinic Type

Approach

Create a binary missingness indicator for hemoglobin and test whether it is associated with clinic type (rural vs. urban).

Contingency table

Clinic	Observed	Missing	Total
Rural (0)	1,842	795	2,637
Urban (1)	1,763	599	2,362
Total	**3,605**	**1,394**	**4,999**

Missing proportions

Rural: 30.15% Urban: 25.36%

Hypotheses

- H_0: Clinic type and hemoglobin missingness are *independent*.
- H_1: Clinic type and hemoglobin missingness are *associated*.

Visual Evidence

A bar plot of missing proportions shows a visible gap: rural clinics have $\sim$5 percentage points more missing hemoglobin than urban clinics. This motivates the formal chi-square test.

MAR: Chi-Square Test of Independence

Test results

- Chi-square statistic: $\chi^2 = 13.967$
- Degrees of freedom: $df = 1$
- p-value: $p = 0.000186$
- Significance level: $\alpha = 0.05$

Decision

Since $p = 0.000186 < 0.05$, we **reject** H_0.

Conclusion
There is a statistically significant association between clinic type and hemoglobin missingness. The proportion of missing hemoglobin differs between rural and urban clinics.

Link to MAR

Since clinic type is an *observed* variable, and missingness depends on it, this finding is **consistent with the MAR mechanism**. The missingness is systematic but explainable by data we already have.

Formal Report

A chi-square test of independence showed a statistically significant association between clinic type and hemoglobin missingness, $\chi^2(1) = 13.97$, $p < .001$, *suggesting that hemoglobin missingness depends on the observed clinic type, consistent with the MAR assumption.*

MAR: Fisher's Exact Test (Robustness Check)

Why Fisher's exact test?
When expected cell frequencies in a contingency table are small, the chi-square approximation may be unreliable. Fisher's exact test provides an exact p-value regardless of sample size.

Results

- Odds Ratio: $OR = 0.787$
- p-value: $p = 0.000170$

Decision
Since $p < 0.05$, reject H_0. The association is confirmed.

Interpretation

The odds ratio of 0.787 indicates that urban clinics have *lower* odds of hemoglobin missingness compared to rural clinics. Both tests agree: missingness is associated with clinic type.

Formal Report

Fisher's exact test indicated a statistically significant association between clinic type and hemoglobin missingness ($OR = 0.787$, $p < .001$), consistent with the MAR assumption.

Note

In this dataset, both chi-square and Fisher's test reach the same conclusion. Fisher's is preferred when any expected count < 5.

MAR: Combined Interpretation & Proportions

Proportions of missing hemoglobin

Clinic Type	Proportion Missing	Percentage
Rural (`urban.clinic = 0`)	0.3015	30.15%
Urban (`urban.clinic = 1`)	0.2536	25.36%

Difference: Rural clinics have ∼4.8 percentage points more missing hemoglobin.

Combined Report

A chi-square test showed a significant association between clinic type and hemoglobin missingness ($\chi^2(1) = 13.97$, $p < .001$). The proportion missing was 30.15% in rural vs. 25.36% in urban clinics. This provides evidence consistent with a MAR mechanism.

Implication for Imputation

Since missingness is explained by clinic type (observed), we can condition on it. Methods like MICE and KNN, which use observed covariates, are appropriate.

What MAR Does NOT Mean

MAR does not mean the *missing hemoglobin values* are random. It means the *probability of being missing* is explained by observed clinic type. The actual unobserved values may still differ across groups.

MNAR: Missing Not At Random

Definition

The probability that a value is missing depends on the *unobserved (missing) value itself*.

Key properties

- The most problematic mechanism — cannot be verified from observed data alone.
- Requires domain expertise, clinical reasoning, and sensitivity analysis.
- Standard imputation methods (MICE, KNN) may produce biased results.
- The missingness mechanism must be explicitly modelled.

Why it matters

If the sickest patients are the ones with missing data, imputing their values from healthier patients *understates* disease severity.

MOH Example

Creatinine (`log_creat_centered`) measurements are missing because patients with severely failing kidneys — and therefore potentially *very high* creatinine — were too critically ill to be transported to the central MOH laboratory.

Another Reason

Some critically ill patients may have *died* before the creatinine measurement could be taken. The missingness is directly tied to the severity of the unobserved value.

MNAR: The Statistical Challenge

Why MNAR cannot be proven

- The values causing the missingness are, by definition, *unobserved*.
- No statistical test on observed data can confirm MNAR.
- A non-significant association with observed variables does *not* automatically prove MNAR.

Assessment approach

- **Clinical knowledge:** Do physicians believe the sickest patients miss labs?
- **Domain expertise:** Are there logistical barriers (transport, mortality)?
- **Sensitivity analysis:** Compare results under different missingness assumptions.
- **Pattern inspection:** Do patients with multiple missing labs have worse outcomes?

Trainer Note

A statistically non-significant association with observed variables does *not* automatically prove MNAR. It simply means you cannot explain the missingness with the variables you checked. MNAR remains a *clinical judgement*.

Key Point

MNAR is the most difficult mechanism to identify because it depends on information that is, by definition, not observed. It requires assumptions that must be stated transparently.

MNAR: Treatment & Reporting Obligations

Treatment options

- **Sensitivity analysis:** Fit models under MCAR, MAR, and MNAR assumptions; compare conclusions.
- **Selection models:** Jointly model the outcome and the missingness process.
- **Pattern-mixture models:** Stratify by missingness pattern and model each subgroup.
- **Expert elicitation:** Consult clinicians about plausible ranges for missing values.

Reporting obligation
Always flag suspected MNAR variables in any report. State the assumption, the clinical reasoning, and the potential direction of bias.

Policy Danger

If creatinine is MNAR and we impute with the median of healthier patients, we *understate* kidney disease severity. Resource allocation models would then *under-prioritise* the facilities serving the sickest patients.

Practical Guidance

- Never silently impute suspected MNAR data.
- Create a missingness indicator as an additional feature.
- Report results with and without imputation.
- Quantify the plausible range of bias.

Missingness Mechanisms: Summary & Classification

Variable	Missing %	Test Used	Result	Mechanism
BMI	13.4%	t-test (age)	$p = 0.352$	MCAR (plausible)
Hemoglobin	27.9%	χ^2 / Fisher	$p < 0.001$	MAR (supported)
Creatinine	30.8%	Clinical reasoning	—	MNAR (suspected)
HIV status	39.2%	Clinical reasoning	—	MNAR (suspected)
Married	3.3%	Low missingness	—	Likely MCAR

Key Insight

Different variables in the *same* dataset can have different missingness mechanisms. Each requires its own assessment and tailored treatment.

Decision Rule

MCAR $\rightarrow$ deletion acceptable. MAR $\rightarrow$ model-based imputation (KNN, MICE). MNAR $\rightarrow$ sensitivity analysis + domain model + flag in report.

Part 3: Preventing Data Leakage

"Split before you impute — or your model is lying to you."

The Critical Rule: Split Before Imputation

The rule
Always split the dataset into training and testing sets *before* performing any imputation.

Why?

- The training set is used to *learn* imputation parameters.
- The test set must remain *unseen* until final evaluation.
- If you impute using the full dataset first, test-set information leaks into training.

Consequence of violation

- Overly optimistic performance estimates.
- Model appears better than it truly is.
- Poor generalisation to new, unseen data.

Data Leakage Defined

Data leakage occurs when information from the test set influences the training process. This inflates performance metrics and produces models that fail in production.

Concrete Example

Training data mean age = 45 years. Test data mean age = 60 years. If you compute the overall mean *before* splitting, the test-set age contaminates the imputation value. Instead: learn 45 from training → apply 45 to both sets.

The Correct Workflow: 7 Steps

Always follow this sequence:

Step	Action	Detail
1	Load the data	Read the raw dataset.
2	Separate X and y	Define predictors and outcome.
3	Split into Train / Test	Stratified split to preserve outcome rates.
4	Fit imputation on Train	Learn parameters (median, KNN distances, MICE models) *only* from training data.
5	Transform Train & Test	Apply learned parameters to both sets.
6	Train the model	Fit the predictive model on imputed training data.
7	Evaluate on Test	Assess performance on imputed test data.

Key Principle

Never allow information from the test set to influence the training process. This applies to imputation, scaling, feature selection, outlier treatment, encoding, dimensionality reduction, and feature engineering.

Best Practice: Use a Pipeline

Why use a pipeline?

- Encapsulates all preprocessing steps in a single object.
- Ensures fit/transform order is always correct.
- Prevents accidental data leakage.
- Makes the workflow reproducible and auditable.

What belongs in the pipeline?

- Imputation (median, KNN, MICE)
- Standardization / scaling
- Feature selection
- Outlier treatment
- Encoding
- Dimensionality reduction

The Mantra

Split → Fit on Train → Transform Train & Test → Train Model → Evaluate on Test

In This Session

We fit each imputer (Median, KNN, MICE) on the training set only, then applied the fitted transformer to both training and test sets. This ensures no test-set information leaks into the imputation parameters.

Common Mistake

Running `fit_transform` on the full dataset *before* splitting. This is the single most common source of data leakage in beginner ML workflows.

Part 4: Imputation Strategies Compared

"Four strategies, one prediction task — let the evidence decide."

Experimental Setup

Prediction task
Classify whether a patient has SBP $\geq$ 120 mmHg (`SBP_ge120`).

Features (5)

- Age
- DBP (diastolic blood pressure)
- BMI
- Hemoglobin (centered)
- Log creatinine (centered)

Train/Test split

- 70% training (3,499 records)
- 30% testing (1,500 records)
- Stratified to preserve outcome rates

Model
Logistic Regression (unpenalized, max_iter = 1000).

Evaluation metrics

- **Accuracy:** proportion correctly classified.
- **AUC-ROC:** ability to discriminate between classes (more informative for imbalanced data).

Why Logistic Regression?

Simple, interpretable, and sensitive to data quality differences. If imputation quality matters, it will show here.

Strategy 1: Listwise Deletion (Complete Case Analysis)

Concept

Remove every row that has *any* missing value in the analysis variables. Only complete cases are retained.

Impact on sample size

- Training: 3,499 → 2,030 (lost 42%)
- Testing: 1,500 → 858 (lost 43%)
- Total retained: 2,888 out of 4,999

When appropriate

- Data are MCAR.
- Missingness is small (<5%).
- Sample size is large enough to absorb the loss.

Results

- Accuracy: **0.9510** (highest)
- AUC-ROC: **0.7516** (lowest)
- Age coefficient: 0.0610

The Paradox

Highest accuracy but lowest AUC. The model looks accurate because the remaining complete cases are a "clean" subset — but it has poor discrimination and wasted 42% of the data.

Bias Risk

If data are NOT MCAR, deletion introduces selection bias. The complete cases may not represent the full population.

Strategy 2: Median Imputation

Concept

Replace every missing value with the *median* of the observed values for that variable (computed from training data only).

Advantages

- Simple and computationally fast.
- Robust to outliers (unlike mean imputation).
- Retains the full sample size.
- Easy to implement and explain.

Limitations

- Treats imputed values as truly observed.
- Artificially reduces variance.
- Distorts correlations between variables.
- Underestimates standard errors.

Results

- Accuracy: **0.9487**
- AUC-ROC: **0.7664**
- Age coefficient: 0.0582
- Sample retained: 3,499 train / 1,500 test (full)

Interpretation

Median imputation preserves the full sample and achieves reasonable discrimination. However, it collapses the distribution of imputed variables toward a single value, weakening multivariate relationships.

Best Use

Exploratory analysis or when missingness is very small ($<5\%$). Not recommended for formal inference.

Strategy 3: KNN Imputation

Concept

For each patient with a missing value, find the k most *similar* patients (based on other observed features) and use their values to estimate the missing one.

How it works

- Compute distances between patients using observed features.
- Identify the $k = 5$ nearest neighbours.
- Impute the missing value as the (weighted) average of neighbours' values.

Advantages

- Uses local patterns and patient similarity.
- Preserves multivariate relationships better than median.
- Adapts to the data structure.

Results

- Accuracy: **0.9493**
- AUC-ROC: **0.7718** (highest)
- Age coefficient: 0.0587
- Sample retained: full (3,499 / 1,500)

Why It Works Here

Patients with similar age, DBP, and BMI are likely to have similar hemoglobin and creatinine levels. KNN exploits this clinical similarity.

Limitations

Computationally expensive for very large datasets (national MOH databases). Requires appropriate variable scaling. Performance depends on the choice of

Strategy 4: MICE (Multivariate Imputation by Chained Equations)

Concept
Iteratively model each variable with missing values as a function of all other variables. Cycle through variables until estimates converge.

How it works

- Initialize missing values (e.g., with median).
- For each incomplete variable, regress it on all others.
- Draw imputed values from the predicted distribution.
- Repeat for 10 iterations until convergence.

Advantages

- Preserves correlations and multivariate structure.
- Appropriate under MAR.
- Can incorporate uncertainty via multiple imputed datasets.

Results

- Accuracy: **0.9487**
- AUC-ROC: **0.7686**
- Age coefficient: 0.0591
- Sample retained: full (3,499 / 1,500)

Why MICE Matters

MICE accounts for the *uncertainty* introduced by imputation. In a full multiple-imputation framework, you generate m imputed datasets, analyse each, and pool results — producing valid standard errors and confidence intervals.

Best Use

Statistical inference with MAR clinical data.
Epidemiological research where valid confidence

Model Performance: Side-by-Side Results

Strategy	Train N	Test N	Accuracy	AUC-ROC	Age Coef.
1. Listwise Deletion	2,030	858	0.9510	0.7516	0.0610
2. Median Imputation	3,499	1,500	0.9487	0.7664	0.0582
3. KNN Imputation	3,499	1,500	0.9493	**0.7718**	0.0587
4. MICE Imputation	3,499	1,500	0.9487	0.7686	0.0591

Key Observations

- KNN achieves the **highest AUC** (0.7718).
- Listwise deletion has the **highest accuracy** but **lowest AUC**.
- All imputation methods retain the full sample.
- Age coefficients are stable across methods ($\sim$0.058–0.061).

Accuracy vs. AUC

Accuracy can be misleading with imbalanced classes. AUC measures the model's ability to *rank* patients correctly — a more informative metric for clinical decision support.

Why KNN Imputation Won

Five reasons KNN is the best overall approach

1. **Highest AUC:** 0.7718 — best discrimination between SBP ≥ 120 and < 120.
2. **High accuracy:** 94.93%, only marginally below listwise deletion.
3. **Full sample retained:** 3,499 training records vs. only 2,030 for deletion.
4. **Better discrimination:** AUC is more informative than accuracy for clinical classification.
5. **Stable coefficients:** Age effect ($\beta = 0.0587$) is consistent across all methods.

Clinical Interpretation

KNN leverages the fact that patients with similar demographics and vitals tend to have similar lab values. This "neighbourhood" logic mirrors clinical reasoning — a 45-year-old with similar DBP and BMI is likely to have similar hemoglobin.

Report Statement

Among the four strategies, KNN imputation provided the best overall predictive performance, achieving the highest AUC (0.772) and a high classification accuracy (94.93%), while retaining the full sample.

Important Caveat: Listwise Deletion

Why we do NOT choose listwise deletion

- Highest accuracy (95.10%) but **lowest AUC** (0.7516).
- Discards 42% of training data and 43% of test data.
- Substantial loss of statistical power.
- If data are not truly MCAR, deletion introduces **selection bias**.
- The "complete" subset may not represent the full patient population.

When deletion IS acceptable

- Data are confirmed MCAR.
- Missingness is very small (<5%).
- Sample size is large enough to absorb the loss without power reduction.

The Accuracy Illusion

High accuracy after deletion can be misleading. By removing patients with missing labs (often the sickest), the remaining sample is "easier" to classify. The model looks good because it never sees the hard cases.

AUC Tells the True Story

AUC of 0.7516 means the model struggles to distinguish between outcome classes. This is hidden by the inflated accuracy. Always report both metrics — and prioritise AUC for clinical models.

Recommended Approach & Narrative Report

For prediction: KNN Imputation

- Highest AUC while retaining the full sample.
- Clinically intuitive: similar patients have similar values.
- Suitable for moderate-sized datasets.

For statistical inference: MICE

- Accounts for imputation uncertainty.
- Produces valid standard errors via multiple imputed datasets.
- Appropriate under MAR — supported by our hemoglobin analysis.

For suspected MNAR variables:

- Sensitivity analysis.
- Missingness indicator as a feature.
- Flag in all reports.

Narrative Report

Among the four missing-data treatment strategies, KNN imputation provided the best overall predictive performance, achieving the highest AUC (0.772) and a high classification accuracy (94.93%). Although listwise deletion yielded a marginally higher accuracy (95.10%), it resulted in substantial loss of observations and had the lowest AUC (0.752). Median and MICE imputation produced slightly lower AUC values of 0.766 and 0.769, respectively. Therefore, KNN imputation was selected as the preferred strategy because it retained the full sample while providing the strongest discriminatory performance.

Best Practices: Method-by-Method Guide

Method	Main Advantage	Main Limitation	Best Use
Deletion	Simple; unbiased under MCAR	Loss of data; selection bias if not MCAR	MCAR + large sample
Mean/Median	Fast; easy to implement	Reduces variance; distorts correlations	Exploratory analysis; small missingness
KNN	Uses patient similarity; preserves local structure	Computationally expensive; needs scaling	Moderate-sized datasets; prediction
MICE	Preserves relationships; accounts for uncertainty	Complex; computationally demanding	MAR clinical data; statistical inference

Universal Rules

- Always split before imputation.
- Fit imputer on training data only.
- Classify the mechanism before choosing a method.

No Universal Best

The appropriate method depends on the missingness mechanism, amount of missingness, research objective, and dataset characteristics.

Key Takeaways

- **Classify before you treat:** MCAR, MAR, and MNAR each demand a different imputation strategy. Misclassification leads to biased estimates.
- **Statistical tests guide classification:** t-tests for MCAR, chi-square/Fisher's for MAR, clinical reasoning for MNAR.
- **Prevent data leakage:** Always split before imputation. Fit preprocessing on training data only. Use pipelines to enforce this.
- **No single method is universally best:** KNN excelled for prediction (AUC $= 0.772$); MICE is preferred for inference; deletion is safe only under MCAR.
- **Accuracy alone is misleading:** AUC-ROC is more informative for clinical classification, especially with imbalanced outcomes.
- **MNAR requires transparency:** Suspected MNAR variables must be flagged, subjected to sensitivity analysis, and reported with explicit assumptions.
- **Every imputation is a modelling decision:** Document what was imputed, how, and under which assumption — this is your audit trail.

Session Summary & What Comes Next

What We Covered Today

- Audited 4,999 MOH patient records for missingness.
- Classified BMI as MCAR, hemoglobin as MAR, creatinine as suspected MNAR.
- Applied the split-before-imputation rule to prevent data leakage.
- Compared 4 imputation strategies via Logistic Regression.
- Recommended KNN for prediction, MICE for inference.

Coming Up in Week 1

- **Next session:** Outlier detection (IQR, Z-score), feature encoding, and scaling/normalisation.
- **Later:** Building reproducible `sklearn` pipelines and handling multi-collinearity.

Preparation

Review the imputation strategy comparison. Think about how you would handle a variable with 60% missingness that you suspect is MNAR. Bring questions.

Q&A & Thank You

Questions?

Let's discuss MCAR vs MAR detection, when KNN outperforms MICE, how to handle suspected MNAR in policy reports, or pipeline design.

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